

Connor Savugot

B.S. in Computer Engineering | North Carolina State University | Graduated May 2026

consav04@gmail.com | +1-828-541-9487 | Murphy, NC | linkedin.com/in/connorsavugot | github.com/Clem085

TECHNICAL PROFILE

Embedded software engineer developing real-time C firmware for power, control, and communications systems across 16- and 32-bit microcontrollers, FreeRTOS, and bare-metal DSPs. Experience spans C++/Python engineering tools, multiprocessor hardware/software integration, datasheet-driven peripheral bring-up, JTAG/bench debugging, and Arm-based Embedded Linux systems.

TECHNICAL SKILLS

Languages: Embedded C, C++, Python, Bash

Embedded & Real-Time: Bare-metal firmware, FreeRTOS, POSIX threads, interrupts/timers, peripheral/HAL bring-up, boot/update/recovery systems

Processors: dsPIC33CK (16-bit DSC), PIC32, TMS320 DSP, STMicroelectronics MCU, MSP430FR2355; TI AM62 Arm-based SoC

Debug & Build: JTAG, register inspection, Code Composer Studio, GCC/Make, compiler/toolchain configuration, Git/GitLab, Docker, oscilloscopes, logic/protocol analyzers

Systems & Interfaces: Embedded Linux, Yocto/Arago, device trees, systemd; PWM, ADC, I2C, SPI, UART/RS-232, CAN 2.0/CAN FD/J1939, IPMI/IPMB

ENGINEERING EXPERIENCE

AEGIS Power Systems

Embedded Firmware Engineer

May 2026–Present

- Develop low-level C firmware for 16-bit dsPIC33CK DSCs and bring up peripherals one subsystem at a time from datasheets, reference manuals, and schematics; isolate register, timing, configuration, and electrical faults using JTAG, targeted tests, oscilloscopes, and logic analyzers.
- Implemented and validated VITA 46.11 Tier 2 management firmware for a VPX power supply, exposing IPMI 2.0 sensor, FRU/inventory, and event data over an I2C-based IPMB link to the chassis manager.
- Integrate battery hardware and CAN-connected power devices with a TI AM62 Arm-based Embedded Linux controller; build Yocto/Arago images and kernels, update device trees, develop external-watchdog services, and trace faults across power, wiring, CAN, boot, configuration, and code.

Embedded Systems Intern Computer Science Intern

May–Aug 2025
May–Aug 2024

Selected work across both internships

- Developed bare-metal C power-control firmware on a TMS320 DSP for transformer-coupled bidirectional buck-boost supplies using 12 PWM A/B pairs (24 physical outputs); configured each A channel's period/frequency, duty cycle, and relative phase while each B output followed or inverted its paired A signal, then validated ADC/PWM behavior in Code Composer Studio.
- Developed embedded C firmware across a two-processor power system: a PIC32 running FreeRTOS coordinated I2C, SPI, CAN/J1939, UART, and update behavior while the bare-metal TMS320 DSP executed real-time converter control.
- Built a resilient update/recovery workflow: a C++ CLI accepted HEX images, a Python TCP sender handled chunking and checksums, and the PIC32 staged new firmware and a golden image in external SPI flash, programmed and verified the DSP over UART, and controlled boot.
- In 2024, built a restricted SSH/chroot interface and Python/Bash commands for a resource-constrained TI Yocto/Arago power system; reverse-engineered RS-232 commands/checksums for programmable loads and relay controllers and helped assemble the associated test circuit.

SELECTED POWER & REAL-TIME PROJECTS

EV Active Sensor Adapter | Battery Monitoring & CAN

NC State University

- Helped architect a custom-PCB system integrating an STMicroelectronics MCU, TI BQ-family battery monitor/ADC, CAN transceivers, power/sensing interfaces, embedded control, and telemetry.
- Translated electrical and protocol constraints plus component trade studies into interface requirements, validation criteria, and staged subsystem tests; coordinated risks, design reviews, and documentation across the team.

Real-Time Systems & Formal Verification | C, Linux, Lustre

Spring 2026

- Implemented and instrumented periodic POSIX-thread tasks in C using CPU affinity, absolute-time releases, priority-inheritance mutexes, and timestamp traces; analyzed EDF plus rate- and deadline-monotonic schedulability and plotted execution timelines.
- Modeled synchronous control logic in Lustre and used bounded model checking and counterexample traces to correct temporal monitors and verify required system invariants.

Software Performance & Machine Learning | C++, Python

Selected Coursework

- Implemented BFS, DFS, and greedy best-first maze solvers in C++ and compared runtime and memory tradeoffs across maze configurations.
- Developed a Python mentor-matching pipeline using sentence-transformer embeddings and configurable weighted scoring; added CSV ingestion, persistent constraints/state, and explainable score output.

TECHNICAL LEADERSHIP

ECE 306 Embedded Systems | Teaching Assistant & Course Mentor

Spring 2025–Spring 2026

- Delivered supplemental lectures and debugged embedded C/MSP430FR2355 firmware and custom-PCB projects across three semesters, diagnosing register, timing, peripheral, toolchain, wiring, and soldering failures across diverse student implementations.